Week 1: Econ 672 Introduction

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Overview

Welcome to Econ 672. We will cover how to implement Stata commands for econometrics, along with data management. Data management will be key for transparent and reproducible work or research. For tonight, we revisit the following topics before delving into our toolbox of methods:

- Julian Reif's Coding Guide Folder Structure
- Executing do-files and making log files
- Automate Graphs
- Helpful Advanced Techniques
- Publish your results

Chapter 1

Coding Guide for Folder Structure

Please see Julian Reif's coding guide at https://julianreif.com/guide/. These are very good tips very about repliciability and organization. Having your files organized in a folder as an example below will not only with organization, but with replication, as well. You can hand over the entire folder structure, and another users will be able to run the master do file with just changing the main folder location once.

Folder Structure

- analysis/
  - data/
  - processed/
  - results/
    - figures/
    - tables/
    - scripts/
      - 1_process_raw_data.do
      - 2_...
  - run.do
- paper/
  - manuscript.tex
  - figures/
  - tables/

Chapter 2

Executing do-files and making log files

From Mitchell 10.3, you should always use a do file when you work with Stata. I find log files helpful, but do files are essential. Do files provide transparency to show your work, and it help provide replication. One key feature of do files are comments. Comments are an essential part for two reasons. It will help yourself when you go back to your own code to describe what is going one. Second, it helps with replication, since someone should be able to run your code top to bottom and get the same results that you have in your paper, report, or document.

Never use point and click. It is available in Stata, but don't lower yourself to a SPSS standard.

In case you are new to do files. We'll start off with a simple example. We'll pull some data, summarize some data, and tabulate some data.

```
use wws.dta, clear
summarize age wage hours
tabulate married
```

```
/Users/Sam/Desktop/Econ 645/Data/Mitchell

(Working Women Survey)

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
         age |      2,246    36.25111    5.437983         21         83
        wage |      2,246    288.2885    9595.692          0     380000
       hours |      2,242    37.21811    10.50914          1         80
```

```
    married |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |        804       35.80       35.80
          1 |      1,442       64.20      100.00
------------+-----------------------------------
      Total |      2,246      100.00
```

We can look at a similar file called example1.do

```
    type example1.do
```

We can actually run a do file within a do file

```
do example1.do
```

```
(Working Women Survey w/fixes)

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
         age |      2,246    36.22707    5.337859         21         48
        wage |      2,244    7.796781     5.82459          0   40.74659
       hours |      2,242    37.21811    10.50914          1         80

    married |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |        804       35.80       35.80
          1 |      1,442       64.20      100.00
------------+-----------------------------------
      Total |      2,246      100.00
```

You can use the doedit command if you want to open the Do-file Editor, but you can just click to open the Do-File (this is an exception from what I said above).

Since we already have the Do-File Editor open, running the doedit will just open a new blank do file.

```
doedit
```

Note: you can write do files in Notepad or TextEdit, but you don't get any of the benefits, such as highlighted text.

Let's look at a do file that uses the log command

```
type example2.do
```

Let's run it.

```
log using example2
use wws2, clear
summarize age wage hours
tabulate married
log close example2
```

Note: It is important to note that when opening a log it must end with a "log close" command. If your do files bombs out (fails to finish) before reaching the log close command, your log will remain open.

Even if we close the log file, we'll get an error if we try to run the do file again. So we need to add the replace option.

```
log using example2, replace
use wws2, clear
summarize age wage hours
tabulate married
log close
```

```
      name:  <unnamed>
       log:  /Users/Sam/Desktop/Econ 645/Data/Mitchell/example2.smcl
  log type:  smcl
 opened on:  19 Jan 2026, 17:01:32

(Working Women Survey w/fixes)

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
         age |      2,246    36.22707    5.337859         21         48
        wage |      2,244    7.796781     5.82459          0   40.74659
       hours |      2,242    37.21811    10.50914          1         80

    married |      Freq.     Percent        Cum.
------------+-----------------------------------
          0 |        804       35.80       35.80
          1 |      1,442       64.20      100.00
------------+-----------------------------------
      Total |      2,246      100.00

      name:  <unnamed>
```

```
      log:  /Users/Sam/Desktop/Econ 645/Data/Mitchell/example2.smcl
 log type:  smcl
closed on:  19 Jan 2026, 17:01:32
-------------------------------------------------------------------------------
```

Chapter 3

Automate Graphs and Figures

Stata makes saving graphs easy, so if there is a minor fix all you need to do is rerun the scripts and have your graph fixed. Use the **graph export** command to save a graph.

```
*Get CPS Data
use jun25pub.dta
gen earnings=pternwa/100 if prerelg==1
histogram earnings if prerelg==1, title("Wages in June 2025") caption("Source: Current Populatio
graph export "Wages_June_2025.png"
```

```
/Users/Sam/Desktop/Econ 645/Data/CPS


(28,617 missing values generated)

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
    earnings |      9,939    1596.086    2184.658          0   13145.81

(bin=39, start=0, width=337.07205)
option replace not allowed
r(198);

r(198);
```

Figure 3.1: Wages Histogram

Chapter 4

Stata Programming Techniques

A brief overview of techniques that will be helpful for completing assignments and empirical project topics include - Local Macros, Loops, Tempfiles, By Sort and Egen, and Weights<

1. Macros
2. Local Macros
3. Global Macros
4. Loops
5. Dataframes and Tempfiles
6. Appending

7. Bysort and EGEN
8. Survey Weights
9. Accessing results in Stata's stored memory

4.1 Stata Programming: Macros

One of the essential parts of Stata programming are macros. Macros variables have many uses and can contain strings or numerics. We can use them in loops, in regressions, in calling temporary files, etc. There are two kinds of macros: local and global.

4.2 Local Macros

Local macros work within a single executed do file and removed once the session is run. If you are running local macros you must run the line of code that initializes the macro and the command that utilizes it.

You need to initialize the local macro E.g.: local i = 1 *Key*: You need to call the macro with `key (or left quote key just above tab key) and ' key. E.g.: display` "i' " *E.g.: replace x = 0 if y = 'i'

```
local i = 1
display "`i'"
```

```
1
```

```
local k = `i' + 1
display "`k'"
```

```
2
```

4.3 Global Macros

Global macros are variables that stays within memory even after a new session is run. These macro varibles can work across multiple do files, as well. You need to initialize a global macro E.g.:

```
global j = 2
```

Next, you call the macro with $

```
display "$j"
```

```
2
```

You can use local and global macro variables for qualifiers, as well

```
replace z = 1 if y = $i
```

Another great use of macros is to find a list of for local levels of a categorical variable, and store it in macro variable

```
levelsof varname, local(levels)
foreach l of local levels {
  command if varname == `l'
}
```

We'll demo this later, but it is quite useful

You can also use local macros to test regression models We'll demo this later, as well

4.4 Looping

Looping has many uses when you want to apply a function or command over a repeated set of values *There is forvalues and foreach - I typically use foreach given it's versatility* but Stata says that forvalues can be more efficient *E.g.: Loop over years and iterate i

4.4.1 Loop over a number

```
local i = 0
foreach num of numlist 2015/2019 {
  display "`num'"
  local i = `i'+1
  display "This is the `i' loop"
}
```

```
2015
This is the 1 loop
2016
This is the 2 loop
2017
This is the 3 loop
2018
This is the 4 loop
2019
This is the 5 loop
```

4.4.2 Loop over a local macro

*E.g.: Loop over months and years to read in new files

```
local month jan feb mar apr may jun jul aug sep oct nov dec
foreach y of numlist 2018/2019 {
  foreach m of local month {
    local filename = "`m'`y'.dta"
    display "`filename'"
  }
}
```

```
jan2018.dta
feb2018.dta
mar2018.dta
apr2018.dta
may2018.dta
jun2018.dta
jul2018.dta
aug2018.dta
sep2018.dta
oct2018.dta
nov2018.dta
dec2018.dta
jan2019.dta
feb2019.dta
mar2019.dta
apr2019.dta
may2019.dta
jun2019.dta
jul2019.dta
aug2019.dta
sep2019.dta
oct2019.dta
nov2019.dta
dec2019.dta
```

4.5 Data Frames and Tempfiles

Data frames are a relatively new part of Stata (Stata 16+) that let you switch between datasets. Before you could only work on one dataset at a time. You can read an introduction to Stata data frames from Asjad Naqvi.

For older versions of Stata, we have the tempfiles workaround. Tempfiles are useful since they are a bypass around an older Stata limitation of one dataframe at a time. Later iterations of Stata introduced the dataframe functionality, but the tempfile method still works well. We will append three years of CPS MORG data from NBER and generate short 1-year panels. *Note*: Tempfiles are really just macros.

4.6 Appending multiple CPS files

Get CPS Data

Small CPS Files have only a few variables in them compared to the usual. The full Census CPS micro data files found at: https://www.census.gov/data/datasets/time-series/demo/cps/cps-basic.html.

Set up initial tempfile so we can add each year of CPS data. Save the macro and set it to emptyok.

```
tempfile cps
save `cps', emptyok
```

Create a local variable to loop over each year and month to append monthly CPS files into 1 cps file.

```
cd "/Users/Sam/Desktop/Data/CPS"
*Set month macro
local month jan feb mar apr may jun jul aug sep oct nov dec
local filecount = 0

tempfile cps
save `cps', emptyok

foreach y of numlist 23/24 {

  foreach m of local month {

    *Show the year and month
    display "`m'`y'"
    local filename "small`m'`y'pub.dta"
    display "`filename'"

    *Open the monthly data file
    use "`filename'", clear

      *Append monthly data file to cps tempfile
      quietly append using `cps'

      save `cps', replace
      clear
```

```
      *Count the number of monthly files appended
      local filecount = `filecount' + 1
    }
}
*Get the appended CPS File
use `cps'
tab hrmonth hryear4
save "cps23_24.dta", replace
```

```
/Users/Sam/Desktop/Data/CPS




(note: dataset contains 0 observations)
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved

jan23
smalljan23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
feb23
smallfeb23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
mar23
smallmar23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
apr23
smallapr23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
may23
smallmay23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
jun23
smalljun23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
jul23
smalljul23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
aug23
smallaug23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
sep23
smallsep23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
```

```
oct23
smalloct23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
nov23
smallnov23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
dec23
smalldec23pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
jan24
smalljan24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
feb24
smallfeb24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
mar24
smallmar24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
apr24
smallapr24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
may24
smallmay24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
jun24
smalljun24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
jul24
smalljul24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
aug24
smallaug24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
sep24
smallsep24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
oct24
smalloct24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
nov24
smallnov24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
dec24
smalldec24pub.dta
file /var/folders/tz/fwgb6p8923dg6yvb_kt_pw3c0000gn/T//S_72032.000001 saved
```

```
           |        HRYEAR4
   HRMONTH |      2023       2024 |     Total
-----------+----------------------+----------
         1 |   125,982    126,802 |   252,784
         2 |   124,754    126,784 |   251,538
         3 |   124,477    124,581 |   249,058
         4 |   126,798    126,850 |   253,648
         5 |   127,559    126,945 |   254,504
         6 |   127,491    126,112 |   253,603
         7 |   127,424    126,158 |   253,582
         8 |   127,930    127,031 |   254,961
         9 |   126,665    126,590 |   253,255
        10 |   127,941    126,387 |   254,328
        11 |   126,917    126,686 |   253,603
        12 |   126,832    126,743 |   253,575
-----------+----------------------+----------
     Total | 1,520,770  1,517,669 | 3,038,439
```

```
file cps23_24.dta saved
```

CPS Basic Data Dictionary can be found at: CPS Basic Data Dictionary.

Check all years are there

4.7 By Sort and EGEN

Please note that we will get into bysort egen later in the semester. However, bysort: egen is a powerful combination that it is worth bring up now. You can summarize, replace, or create new variables by multiple groups with the by sort and egen commands Let's generate laborforce

1. 1 is employed at work;
2. 2 is employed absent;
3. 3 is unemployed layoff;
4. 4 is unemployed looking;
5. 5 is NILF retired;
6. 6 is NILF disabiled; and
7. 7 is NILF other

```
gen laborforce = .
replace laborforce = 0 if pemlr >= 5 & pemlr <= 7
replace laborforce = 1 if pemlr >= 1 & pemlr <= 4
label define laborforce1 0 "NILF" 1 "Labor Force"
```

```
label values laborforce laborforce1
tab laborforce

gen employed = .
replace employed = 0 if pemlr >= 3 & pemlr <= 7
replace employed = 1 if pemlr >= 1 & pemlr <= 2
label define employed1 0 "Not Employed" 1 "Employed"
label values employed employed1
tab employed
```

Generate a race/ethnicity category from existing

```
gen race_ethnicity = .
replace race_ethnicity = 1 if ptdtrace == 1 & pehspnon == 2
replace race_ethnicity = 2 if ptdtrace == 2 & pehspnon == 2
replace race_ethnicity = 3 if pehspnon == 1
replace race_ethnicity = 4 if ptdtrace == 3 & pehspnon == 2
replace race_ethnicity = 5 if (ptdtrace == 4 | ptdtrace == 5) & pehspnon == 2
replace race_ethnicity = 6 if (ptdtrace >= 6 & ptdtrace <= 26) & pehspnon == 2
label define race_ethnicity1 1 "White NH" 2 "Black NH" 3 "Hispanic or Latino/a" ///
4 "Native American NH" 5 "Asian or Pacific Islander NH" 6 "Multiracial NH"
label values race_ethnicity race_ethnicity1
tab race_ethnicity
```

Sort by Sex

```
sort pesex
*Summarize laborforce by sex
by pesex: sum laborforce
bysort: sum laborforce
```

Sort by Sex and Race.

```
sort pesex race_ethnicity
*Summarize laborforce by sex and race
by pesex race_ethnicity: sum laborforce
bysort pesex race_ethnicity: sum laborforce
```

Generate Age Bin to find individuals over 16.

```
gen over_16 = .
replace over_16 = 0 if prtage < 16
replace over_16 = 1 if prtage >= 16
```

```
label define over_16a 0 "Under 16" 1 "16 and older"
label values over_16 over_16a
tab over_16
```

Generate unweighted laborforce participation rate for each group with by sort and egen. Bysort works, as well, but I usually use sort on one line and by on the other

```
sort pesex race_ethnicity
by pesex race_ethnicity: egen mean_lfpr = mean(laborforce) if over_16 == 1
by pesex race_ethnicity: sum mean_lfpr
```

```
*Counting and indexing/subscripting within groups
gen idcount = .
by pesex race_ethnicity: replace idcount = _n
gen idcount2 = .
by pesex race_ethnicity: replace idcount2 = idcount[_N]
```

4.8 Retrieving estimates stored in Stata's memory

Instead manually entering summary or regression results, we can use scalars in Stata's memory to retrieve the information. This reduces human error of copying and pasting.

For summary statistics we can use the **return list** command.

```
cd "/Users/Sam/Desktop/Data/CPS"
use smalljan24pub.dta, clear
sum pternwa if prerelg==1

return list

gen demean_earnings=pternwa-`r(mean)' if prerelg==1
summarize demean_earnings
```

```
/Users/Sam/Desktop/Data/CPS

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
     pternwa |     10,666    137274.8    146839.5          0    1125627
```

```
scalars:
                  r(N) =  10666
              r(sum_w) =  10666
               r(mean) =  137274.7503281455
                r(Var) =  21561833661.67427
                 r(sd) =  146839.4826389492
                r(min) =  0
                r(max) =  1125627
                r(sum) =  1464172487

(116,136 missing values generated)

    Variable |        Obs        Mean    Std. Dev.       Min        Max
-------------+---------------------------------------------------------
demean_ear~s |     10,666   -3.79e-12    146839.5  -137274.8   988352.2
```

We can also grab information about a regression

```
cd "/Users/Sam/Desktop/Data/CPS"
use smalljan24pub.dta, clear
gen lnearn=ln(pternwa) if prerelg==1
gen exp=prtage-16
replace exp=0 if exp==-1

reg lnearn c.exp##c.exp i.peeduca i.peernlab if prerelg==1

ereturn list

matrix list e(b)

*Coefficients and standard errors
display _b[1.peernlab] _se[1.peernlab]
```

```
/Users/Sam/Desktop/Data/CPS


(116,150 missing values generated)

(27,667 missing values generated)

(1,209 real changes made)

      Source |       SS           df       MS      Number of obs   =    10,652
-------------+----------------------------------   F(18, 10633)    =    238.75
```

```
       Model |  2310.98949        18  128.388305   Prob > F        =    0.0000
    Residual |  5717.80316    10,633  .537741292   R-squared       =    0.2878
-------------+----------------------------------   Adj R-squared   =    0.2866
       Total |  8028.79265    10,651  .753806465   Root MSE        =    .73331

------------------------------------------------------------------------------
      lnearn |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
-------------+----------------------------------------------------------------
         exp |   .0654986   .0019087    34.31   0.000     .0617571    .0692401
             |
 c.exp#c.exp |  -.0010166   .0000324   -31.38   0.000    -.0010801   -.0009531
             |
     peeduca |
         32  |   .2151811   .2348611     0.92   0.360    -.2451906    .6755528
         33  |   .1297205   .2243499     0.58   0.563    -.3100474    .5694884
         34  |   .1947927   .2240222     0.87   0.385    -.2443328    .6339181
         35  |  -.0479483   .2153955    -0.22   0.824    -.4701637    .3742672
         36  |  -.1032521   .2123809    -0.49   0.627    -.5195583    .3130541
         37  |   .0602974   .2103972     0.29   0.774    -.3521206    .4727154
         38  |   .1407128   .2128599     0.66   0.509    -.2765325    .5579581
         39  |   .4391957    .203923     2.15   0.031     .0394684     .838923
         40  |   .4534936   .2043117     2.22   0.026     .0530045    .8539827
         41  |   .5707234   .2059659     2.77   0.006     .1669917     .974455
         42  |   .5189681   .2053085     2.53   0.011     .1165251    .9214112
         43  |    .914999   .2038914     4.49   0.000     .5153337    1.314664
         44  |   1.025085   .2044413     5.01   0.000     .6243415    1.425828
         45  |   1.463769   .2108507     6.94   0.000     1.050462    1.877076
         46  |     1.3309   .2076626     6.41   0.000     .9238427    1.737958
             |
  2.peernlab |  -.0496265    .024125    -2.06   0.040    -.0969159   -.0023371
       _cons |   10.07725   .2065816    48.78   0.000     9.672309    10.48219
------------------------------------------------------------------------------


scalars:
                  e(N) =  10652
               e(df_m) =  18
               e(df_r) =  10633
                  e(F) =  238.7547824846974
                 e(r2) =  .2878377352595289
               e(rmse) =  .7333084563678371
                e(mss) =  2310.989494457061
                e(rss) =  5717.803159756108
               e(r2_a) =  .2866321563292809
                 e(ll) =  -11800.89312136709
               e(ll_0) =  -13608.80112438194
```

```
               e(rank) =  19

macros:
            e(cmdline) : "regress lnearn c.exp##c.exp i.peeduca i.peernlab if prerelg==1"
              e(title) : "Linear regression"
          e(marginsok) : "XB default"
                e(vce) : "ols"
             e(depvar) : "lnearn"
                e(cmd) : "regress"
         e(properties) : "b V"
            e(predict) : "regres_p"
          e(estat_cmd) : "regress_estat"

matrices:
                  e(b) :  1 x 21
                  e(V) :  21 x 21

functions:
             e(sample)


e(b)[1,21]
                    c.exp#        31b.         32.         33.         34.         35.         3
           exp       c.exp     peeduca     peeduca     peeduca     peeduca     peeduca     peeduc
y1   .06549863  -.00101659           0   .21518108   .12972052   .19479266  -.04794826   -.103252

            38.         39.         40.         41.         42.         43.         44.         4
       peeduca     peeduca     peeduca     peeduca     peeduca     peeduca     peeduca     peeduc
y1    .1407128   .43919568    .4534936   .57072336   .51896812   .91499896   1.0250848    1.46376

            1b.          2.
      peernlab    peernlab       _cons
y1           0  -.04962649   10.077247

00
```

4.9 Survey Weights

We can use the Current Population Survey public use microdata and replicate published BLS numbers. We need to adjust our numbers with the appropriate weights in order to replicate the published numbers.

First, we need to adjust weights the weights. In the CSV files the weights need to be divided by 1000 since the documentation says implies 4 decimals.

Composite weights *cmpwgt2** are used for final BLS tabulations of labor force

```
gen cmpwgt2 = pwcmpwgt/10000
```

Outgoing Rotation Weights *pworwgt* are used for earnings only person in interviews 4 and 8

```
gen orwgt2 = pworwgt/10000
```

PWSSWGT weights are used for general tabulations of sex, race, states, etc.

```
gen sswgt2 = pwsswgt/10000
```

PWVETWGT are used to study the veteran population

```
gen vetwgt2 = pwvetwgt/10000
```

PWLWGT are weights used to study someone over multiple CPS interviews

```
gen lgwgt2 = pwlgwgt/10000
```

If we are appending all months, we need to divide by 12 for the Basic CPS to get annual weights.

```
    replace cmpwgt2 = cmpwgt2/12
```

If we are appending all months across multiple years, we need to get a composite weight for all of the CPS files

```
cd "/Users/Sam/Desktop/Data/CPS"
use cps23_24.dta
gen cmpwgt3 = pwcmpwgt/24
```

Use the SvySet command to set up the survey design.

```
    svyset [pw=cmpwgt3]
```

Use the svy: command vars to utilize the survey design.

```
    svy: tab employed hryear4, count cellwidth(20) format(%20.2gc)
```

```
cd "/Users/Sam/Desktop/Data/CPS"
use cps23_24.dta
gen cmpwgt3 = pwcmpwgt/10000
replace cmpwgt3 = cmpwgt3/12
svyset [pw=cmpwgt3]
gen employed = .
replace employed = 0 if pemlr >= 3 & pemlr <= 7
replace employed = 1 if pemlr >= 1 & pemlr <= 2
label define employed1 0 "Not Employed" 1 "Employed"
label values employed employed1
tab employed
svy: tab employed hryear4, count cellwidth(20) format(%20.2gc)
```

```
/Users/Sam/Desktop/Data/CPS


(655,995 missing values generated)

(1,946,105 real changes made)

      pweight: cmpwgt3
          VCE: linearized
  Single unit: missing
     Strata 1: <one>
         SU 1: <observations>
        FPC 1: <zero>

(3,038,439 missing values generated)

(838,080 real changes made)

(1,138,576 real changes made)

    employed |      Freq.     Percent        Cum.
-------------+-----------------------------------
Not Employed |    838,080       42.40       42.40
    Employed |  1,138,576       57.60      100.00
-------------+-----------------------------------
       Total |  1,976,656      100.00

(running tabulate on estimation sample)

Number of strata   =         1                Number of obs     =    1,976,656
```

```
Number of PSUs     = 1,976,656              Population size   =  535,513,623
                                            Design df         =    1,976,655

----------------------------------------------------------------------------
          |                            HRYEAR4
 employed |             2023                 2024                 Total
----------+-----------------------------------------------------------------
 Not Empl |      105,905,668          107,225,904           213,131,572
 Employed |      161,036,522          161,345,530           322,382,051
          |
    Total |      266,942,190          268,571,433           535,513,623
----------------------------------------------------------------------------
  Key:  weighted count

  Pearson:
    Uncorrected   chi2(1)          =   12.9838
    Design-based  F(1, 1976655)    =    9.8919     P = 0.0017
```

Chapter 5

Displaying your results with publication-quality tables with estout and outreg

Stata has two suites of functions that help produce publication-quality tables: estout and outreg2. While it is easy to copy and paste tables, this is not a best practice. You want to have your Stata scripts reduce the amounts of pointing-and-clicking. Pointing-and-clicking is highly repeative and more prone to user error. It is better to avoid doing this! Using estout or outreg2 will help you in the long-run.

5.1 Estout

Estout and its functions are a way to produce publication-quality tables of descriptive statistics or regression results. The main function will be *esttab*.

You can install estout by running the following ssc install command: http://repec.sowi.unibe.ch/stata/estout/index.html

```
ssc install estout
```

Esttab is the main function in estout that provides publication-quality tables. The basics of esttab are the following:

```
esttab [ namelist ] [ using filename ] [ , options estout_options ]
```

5.1.1 Esttab Examples

Here is an example of a simple regression output comparing two models. We use eststo to store the model and use esttab to display the results.

```
quietly sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2
```

```
                      (1)             (2)   
                    price           price   
--------------------------------------------
weight              1.747**         3.465***
                   (2.72)          (5.49)   

mpg                -49.51           21.85   
                  (-0.57)          (0.29)   

foreign                            3673.1***
                                   (5.37)   

_cons              1946.1         -5853.7   
                   (0.54)         (-1.73)   
--------------------------------------------
N                      74              74   
--------------------------------------------
t statistics in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

We can replace t-statistics with pvalues with "p". Or, we can use "se" for standard errors, and we can use ci "ci" for confidence interval. We can add scalars to the model output as well. We can use "F" to display the overall F-statistic, "df_r" for degrees of freedom, and "rmse" for root mean squared error. We can also use "r2" for R-Squared and "ar2" for adjusted R-Squared.

```
quietly sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2, p scalars(F df_r rmse) r2 ar2
```

```
                      (1)             (2)   
                    price           price   
--------------------------------------------
```

```
weight              1.747**         3.465***
                  (0.008)         (0.000)

mpg                -49.51           21.85
                  (0.567)         (0.769)

foreign                            3673.1***
                                  (0.000)

_cons              1946.1         -5853.7
                  (0.590)         (0.087)
--------------------------------------------
N                      74              74
R-sq                0.293           0.500
adj. R-sq           0.273           0.478
F                   14.74           23.29
df_r                   71              70
rmse               2514.0          2130.8
--------------------------------------------
p-values in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

With the title() option, we can add a title to the table output. We can also change the model names at the top of the columns with the mtitle() option. We can move the p-values, t-statistics, or standard errors to the right of the beta coefficients with the option "wide". We can also add a note with the addnote() option

```
quietly sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2, title("Regression Analysis Example") p scalars(F df_r rmse) r2 ar2 mtitle("Model 1'
```

```
Regression Analysis Example
----------------------------------------------------------------------
                      (1)                             (2)
                  Model 1                         Model 2
----------------------------------------------------------------------
weight              1.747**         (0.008)         3.465***        (0.000)
mpg                -49.51           (0.567)         21.85           (0.769)
foreign                                            3673.1***        (0.000)
_cons              1946.1           (0.590)       -5853.7           (0.087)
----------------------------------------------------------------------
N                      74                              74
R-sq                0.293                           0.500
```

```
adj. R-sq           0.273                         0.478
F                   14.74                         23.29
df_r                   71                            70
rmse               2514.0                        2130.8
----------------------------------------------------------------------
p-values in parentheses
* p<0.05, ** p<0.01, *** p<0.001
```

5.1.2 Exporting into Word, Excel, or LaTex format

**Do not copy and paste! Automate!*

We can use esttab with the export options to export into an RTF (Word), Excel, or LaTex format.

5.1.2.1 Export into Excel format

If you want to use your table in Excel. Don't copy and paste! Just add "using" after esttab and include the ".csv" file format to the file name

```
sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2 using example.csv, title("Regression Analysis Example") p scalars(F df_r r
```

5.1.2.2 Export into Word format

If you want to add your table to a Word document, add "using" after esttab and include the ".rtf" file format to the file name.

```
sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2 using example.rtf, title("Regression Analysis Example") p scalars(F df_r r
```

5.1.2.3 Export into LaTex format

You can also export your tables into a LaTex table. You can export your file and load it directly into Overleaf. Add "using" after esttab and include the ".tex" file format to the file name

```
sysuse auto
eststo m1: quietly regress price weight mpg
eststo m2: quietly regress price weight mpg foreign
esttab m1 m2 using example.tex, title("Regression Analysis Example") p scalars(F df_r rmse) r2 ar
```

5.2 Outreg2

Outreg2 is another set of functions that help produce publication-quality tables. My preference is for esttab in estout, so I will provide you with a reference to get started with outreg2 if you are interested.

https://www.princeton.edu/~otorres/Outreg2.pdf

```
ssc install outreg2
```

5.3 Example: Mincer Equations - Replicate Tables

Important Note: Missing values are set to -1 in the CPS PUMS (public-use micro dataset). If you do not account for missing, you will have an incorrect analysis.

First, let's generate some mutually exclusive categories of interest. Recategorize Female:

```
gen female = .
replace female = 0 if pesex == 1
replace female = 1 if pesex == 2
label define female1 0 "Male" 1 "Female"
label values female female1
```

Generate Union

```
gen union = .
replace union = 0 if peernlab == 2
replace union = 1 if peernlab == 1
label define union1 0 "Nonunion" 1 "Union"
label values union union1
```

Our outcome of interest: Weekly Earnings - *PTERNWA*. *Note:* Documentation says that the data in pternwa imply 2 decimals, so we need to divide by 100. We will also use the variable PRERELG to flag individuals with valid earnings.

```
gen earnings = .
replace earnings = pternwa if prerelg==1
*Divide by 100 for decimals
replace earnings = earnings/100
```

Generate Educational Bins

```
tab peeduca
gen educ = .
*High School Drop Out: from Less than 1st Grade to 12th Grade No Diploma
replace educ = 1 if peeduca >= 31 & peeduca <38
*Graduated High School or GED
replace educ = 2 if peeduca == 39
*Some College
replace educ = 3 if peeduca == 40
*AA Degree: Vocational or Academic
replace educ = 4 if peeduca == 41 | peeduca == 42
*Bachelor's Degree
replace educ = 5 if peeduca == 43
*Advanced Degree: Master's, Professional, or Doctorate
replace educ = 6 if peeduca >= 44 & peeduca <= 46
label define educ1 1 "High School Dropout" 2 "High School Graduate" ///
                   3 "Some College" 4 "Associates (VorA) Degree" ///
                           5 "Bachelor's Degree" 6 "Advanced Degree"
label values educ educ1
```

Caveat with Generating Categorical Variables in Stata Don't do peeduca >= 44 without peeduca <= 46 since missing values are very large. So you if do peeduca >= 44 and missing educational status is ".", then missing observations will get categorized as an advanced degree which will be a measurement error.

Generate Potential Experience

```
gen exp = prtage - 16
gen exp2 = exp*exp
```

You can use local macros for testing models

```
local rhs1 i.educ exp exp2
local rhs2 i.educ exp exp2 i.female
local rhs3 i.educ exp exp2 i.female i.union
local rhs4 i.educ exp exp2 i.female i.union i.hryear4
local rhs5 i.educ exp exp2 i.female i.union i.hryear4 i.peio1icd
*Add interaction between female and union
local rhs6 i.educ exp exp2 i.female##i.union i.hryear4 i.peio1icd
```

Run our regression

```
reg earnings `rhs1', robust
reg earnings `rhs2', robust
reg earnings `rhs3', robust
reg earnings `rhs4', robust
reg earnings `rhs5', robust
reg earnings `rhs6', robust
```

```
Linear regression                               Number of obs     =    248,811
                                                F(7, 248803)      =   11601.57
                                                Prob > F          =     0.0000
                                                R-squared         =     0.2645
                                                Root MSE          =     .72359

------------------------------------------------------------------------------------------
                          |               Robust
               lnearnings |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
--------------------------+---------------------------------------------------------------
                     educ |
    High School Graduate  |    .409167   .0061648    66.37   0.000     .3970841    .4212499
            Some College  |   .4095462   .0067562    60.62   0.000     .3963043     .422788
Associates (VorA) Degree  |   .5171878   .0069999    73.88   0.000     .5034681    .5309075
       Bachelor's Degree  |   .8685378   .0063983   135.75   0.000     .8559973    .8810782
         Advanced Degree  |   1.073213   .0069293   154.88   0.000     1.059631    1.086794
                          |
                      exp |    .062107   .0004216   147.31   0.000     .0612807    .0629333
                     exp2 |  -.0009771   7.65e-06  -127.80   0.000     -.000992   -.0009621
                    _cons |   5.518003   .0069875   789.70   0.000     5.504308    5.531699
------------------------------------------------------------------------------------------


Linear regression                               Number of obs     =    248,811
                                                F(8, 248802)      =   12342.42
                                                Prob > F          =     0.0000
                                                R-squared         =     0.3031
                                                Root MSE          =     .70438

------------------------------------------------------------------------------------------
                          |               Robust
               lnearnings |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
--------------------------+---------------------------------------------------------------
                     educ |
    High School Graduate  |   .4156047   .0059502    69.85   0.000     .4039425    .4272668
            Some College  |   .4408511   .0065453    67.35   0.000     .4280225    .4536797
```

```
Associates (VorA) Degree  |   .5596453   .0067703    82.66   0.000     .5463758    .572
       Bachelor's Degree  |   .9084709   .0061978   146.58   0.000     .8963234    .920
         Advanced Degree  |   1.121183   .0067401   166.34   0.000     1.107973    1.13
                          |
                      exp |   .0613624   .0004132   148.52   0.000     .0605526    .062
                     exp2 |  -.0009637   7.50e-06  -128.53   0.000    -.0009784    -.00
                          |
                   female |
                  Female  |   -.333084   .0028429  -117.16   0.000     -.338656   -.327
                    _cons |   5.660114   .0069181   818.17   0.000     5.646555    5.67
------------------------------------------------------------------------------------------


Linear regression                               Number of obs     =    248,811
                                                F(9, 248801)      =   11129.25
                                                Prob > F          =     0.0000
                                                R-squared         =     0.3040
                                                Root MSE          =      .7039

------------------------------------------------------------------------------------------
                          |               Robust
               lnearnings |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Inter
--------------------------+---------------------------------------------------------------
                     educ |
    High School Graduate  |   .4125402   .0059457    69.39   0.000     .4008868    .424
            Some College  |   .4372733   .0065413    66.85   0.000     .4244525    .450
Associates (VorA) Degree  |   .5552335   .0067633    82.09   0.000     .5419776    .568
       Bachelor's Degree  |   .9050996   .0061986   146.02   0.000     .8929506    .917
         Advanced Degree  |   1.114006   .0067622   164.74   0.000     1.100752     1.1
                          |
                      exp |   .0609065   .0004135   147.28   0.000      .060096     .06
                     exp2 |  -.0009571   7.50e-06  -127.69   0.000    -.0009718   -.000
                          |
                   female |
                  Female  |  -.3322049   .0028417  -116.91   0.000    -.3377745   -.326
                          |
                    union |
                   Union  |   .0886408   .0041953    21.13   0.000     .0804181    .096
                    _cons |   5.660998   .0069149   818.66   0.000     5.647445    5.67
------------------------------------------------------------------------------------------


Linear regression                               Number of obs     =    248,811
                                                F(10, 248800)     =   10030.61
                                                Prob > F          =     0.0000
                                                R-squared         =     0.3049
```

```
                                                Root MSE          =     .70345

------------------------------------------------------------------------------------------
                          |               Robust
              lnearnings |      Coef.   Std. Err.      t    P>|t|     [95% Conf. Interval]
--------------------------+---------------------------------------------------------------
                     educ |
     High School Graduate  |   .4122138   .0059457    69.33   0.000     .4005605    .4238672
             Some College  |   .4373993   .0065415    66.86   0.000      .424578    .4502205
 Associates (VorA) Degree  |   .5545629   .0067628    82.00   0.000      .541308    .5678177
        Bachelor's Degree  |   .9045909   .0061969   145.97   0.000     .8924451    .9167367
          Advanced Degree  |   1.113443   .0067577   164.77   0.000     1.100198    1.126688
                          |
                      exp |   .0609184   .0004136   147.29   0.000     .0601077    .0617291
                     exp2 |  -.0009573   7.50e-06  -127.69   0.000     -.000972   -.0009426
                          |
                   female |
                   Female  |  -.3321621   .0028397  -116.97   0.000    -.3377278   -.3265963
                          |
                    union |
                    Union  |   .0887879    .004192    21.18   0.000     .0805716    .0970041
                          |
                  hryear4 |
                     2024  |   .0501524   .0028201    17.78   0.000      .044625    .0556798
                          |
                    _cons |   5.636116   .0070549   798.90   0.000     5.622289    5.649943
------------------------------------------------------------------------------------------
```

```
Linear regression                               Number of obs     =    248,811
                                                F(22, 248788)     =    5310.25
                                                Prob > F          =     0.0000
                                                R-squared         =     0.3349
                                                Root MSE          =     .68812

---------------------------------------------------------------------------------------------
                                                |               Robust
                                     lnearnings |      Coef.   Std. Err.      t    P>|t|     [9
-----------------------------------------------+---------------------------------------------
                                           educ |
                           High School Graduate  |   .3840322   .0059144    64.93   0.000     .3
                                   Some College  |   .4091063   .0065253    62.70   0.000      .
                       Associates (VorA) Degree  |   .5233398   .0068026    76.93   0.000      .
                              Bachelor's Degree  |   .8481448   .0063508   133.55   0.000     .8
                                Advanced Degree  |     1.0828   .0069954   154.79   0.000     1.
                                                |
```

```
                                      exp |   .0550617   .0004126   133.46   0
                                     exp2 |  -.0008699   7.42e-06  -117.31   0
                                          |
                                   female |
                                  Female  |  -.2669904   .0030367   -87.92   0
                                          |
                                    union |
                                   Union  |   .1018213   .0042456    23.98   0
                                          |
                                  hryear4 |
                                    2024  |   .0511416   .0027589    18.54   0
                                          |
                                 prmjind1 |
                       NAICS 21: Mining  |   .4601445   .0215585    21.34   0
                 NAICS 23: Construction  |   .2414286   .0139472    17.31   0
             NAICS 31-33: Manufacturing  |   .2237477   .0137168    16.31   0
     NAICS 42,44-45: Wholesale & Retail  |  -.0344484   .0137914    -2.50   0
NAICS 22,48-49: Utilities & Transportation  |   .1267323   .0143908     8.81   0
                 NAICS 51: Information  |   .2449216   .0178831    13.70   0
      NAICS 52-53: Financial Activities  |   .2939349   .0142959    20.56   0
NAICS 53,55,56: Professional & Business Services  |   .2228786   .0138884    16.05   0
        NAICS 61-62: Education & Health  |  -.0262834   .0136592    -1.92   0
     NAICS 71-72: Leisure & Hospitality  |  -.1961534   .0141092   -13.90   0
              NAICS 81: Other Services  |  -.1098199    .015223    -7.21   0
        NAICS 92: Public Administration  |   .1422022   .0142532     9.98   0
                                          |
                                    _cons |   5.638847   .0146119   385.91   0
-------------------------------------------------------------------------------


Linear regression                               Number of obs     =    248,811
                                                F(23, 248787)     =    5081.12
                                                Prob > F          =     0.0000
                                                R-squared         =     0.3349
                                                Root MSE          =     .68812

-------------------------------------------------------------------------------
                                          |               Robust
                               lnearnings |      Coef.   Std. Err.      t    P
------------------------------------------+------------------------------------
                                     educ |
                   High School Graduate  |   .3841487   .0059155    64.94   0
                           Some College  |   .4092612   .0065272    62.70   0
              Associates (VorA) Degree  |   .5235541   .0068043    76.94   0
                     Bachelor's Degree  |   .8481453   .0063508   133.55   0
                       Advanced Degree  |    1.08265    .006998   154.71   0
```

```
                                              |
                                          exp |   .0550613   .0004126   133.46   0.000     .0
                                         exp2 |  -.0008699   7.42e-06  -117.31   0.000    -.0
                                              |
                                       female |
                                      Female  |  -.2680058   .0031915   -83.97   0.000    -.2
                                              |
                                        union |
                                       Union  |   .0964584   .0056709    17.01   0.000     .0
                                              |
                                 female#union |
                                Female#Union  |   .0112795    .008181     1.38   0.168    -.
                                              |
                                      hryear4 |
                                        2024  |   .0511369   .0027589    18.53   0.000     .0
                                              |
                                     prmjind1 |
                           NAICS 21: Mining   |   .4602906   .0215562    21.35   0.000     .
                     NAICS 23: Construction   |   .2417802   .0139496    17.33   0.000     .2
                 NAICS 31-33: Manufacturing   |   .2239984   .0137179    16.33   0.000     .1
         NAICS 42,44-45: Wholesale & Retail   |  -.0342786    .013792    -2.49   0.013    -.0
 NAICS 22,48-49: Utilities & Transportation   |   .1272795   .0143997     8.84   0.000     .0
                      NAICS 51: Information   |   .2452513   .0178858    13.71   0.000     .2
        NAICS 52-53: Financial Activities     |   .2942334   .0142996    20.58   0.000     .2
NAICS 53,55,56: Professional & Business Services | .2231206 .0138903  16.06   0.000     .1
          NAICS 61-62: Education & Health     |    -.02622   .0136591    -1.92   0.055    -.0
         NAICS 71-72: Leisure & Hospitality   |  -.1959203   .0141108   -13.88   0.000    -.2
                   NAICS 81: Other Services   |  -.1095106    .015225    -7.19   0.000    -.1
            NAICS 92: Public Administration   |   .1426441   .0142568    10.01   0.000     .1
                                              |
                                        _cons |   5.639084   .0146121   385.92   0.000     5.
-----------------------------------------------------------------------------------------------
```

Use esttab for formatted results: esttab documentation.

```
local rhs4 i.educ exp exp2 i.female i.union i.hryear4
local rhs5 i.educ exp exp2 i.female i.union i.hryear4 i.peio1icd
*Add interaction between female and union
local rhs6 i.educ exp exp2 i.female##i.union i.hryear4 i.peio1icd
```

Use est clear to clear out your prior estimates.

```
est clear
```

We use eststo to save a model to compare it to another.

```
eststo reg1: reg earnings `rhs4'
eststo reg2: reg earnings `rhs5'
eststo reg3: reg earnings `rhs6'
```

Output the results

```
esttab, title (Mincer Equation) r2 se noconstant star(* .10 ** .05 *** .01) ///
b(%10.3f) drop (*peio1icd) wide label.
```

```
Mincer Equation
------------------------------------------------------------------------------------
                           (1)                     (2)                     (3)
                          OLS1                    OLS2                    OLS3
------------------------------------------------------------------------------------
High School Gr~e     0.412***   (0.006)      0.384***   (0.006)      0.384***   (0.006)
Some College         0.437***   (0.007)      0.409***   (0.007)      0.409***   (0.007)
Associates (Vo~e     0.555***   (0.007)      0.523***   (0.007)      0.524***   (0.007)
Bachelor's Deg~e     0.905***   (0.006)      0.848***   (0.006)      0.848***   (0.006)
Advanced Degree      1.113***   (0.007)      1.083***   (0.007)      1.083***   (0.007)
exp                  0.061***   (0.000)      0.055***   (0.000)      0.055***   (0.000)
exp2                -0.001***   (0.000)     -0.001***   (0.000)     -0.001***   (0.000)
Female              -0.332***   (0.003)     -0.267***   (0.003)     -0.268***   (0.003)
Union                0.089***   (0.005)      0.102***   (0.005)      0.096***   (0.007)
HRYEAR4=2024         0.050***   (0.003)      0.051***   (0.003)      0.051***   (0.003)
NAICS 11: Agri~e                             0.000          (.)      0.000          (.)
NAICS 21: Mining                             0.460***   (0.022)      0.460***   (0.022)
NAICS 23: Cons~n                             0.241***   (0.014)      0.242***   (0.014)
NAICS 31-33: M~g                             0.224***   (0.014)      0.224***   (0.014)
NAICS 42,44-45~a                            -0.034**    (0.014)     -0.034**    (0.014)
NAICS 22,48-49~n                             0.127***   (0.015)      0.127***   (0.015)
NAICS 51: Info~n                             0.245***   (0.017)      0.245***   (0.017)
NAICS 52-53: F~e                             0.294***   (0.014)      0.294***   (0.014)
NAICS 53,55,56~B                             0.223***   (0.014)      0.223***   (0.014)
NAICS 61-62: E~h                            -0.026*     (0.014)     -0.026*     (0.014)
NAICS 71-72: L~i                            -0.196***   (0.014)     -0.196***   (0.014)
NAICS 81: Othe~s                            -0.110***   (0.015)     -0.110***   (0.015)
NAICS 92: Publ~n                             0.142***   (0.015)      0.143***   (0.015)
Female # Union                                                       0.011      (0.009)
Constant             5.636***   (0.007)      5.639***   (0.015)      5.639***   (0.015)
------------------------------------------------------------------------------------
Observations        248811                  248811                  248811
R-squared            0.305                   0.335                   0.335
------------------------------------------------------------------------------------
Standard errors in parentheses
* p<.10, ** p<.05, *** p<.01
```

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